



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 107



**SAKAI 4 x 4 Centrifugal Pump coupled to
YANMAR TF70L Diesel Engine**

SAKAI 4 x 4 Centrifugal Pump coupled to a YANMAR TF70L Diesel Engine

Test Report No.	:	2013 - 107
Date of Test	:	June 5, 2013
Test Valid Until	:	June 5, 2018
Brand	:	SAKAI
Model	:	4 x 4
Serial No.	:	No data
Type and Size	:	Single-end suction, 100 x 100 mm, Centrifugal pump
Manufacturer	:	Made in China
Test Requested by	:	ALL CERTIFIED EQUIPMENT TRADING CORPORATION 905 EDSA PhilAm, Quezon City, Philippines
Machine Classification	:	Commercial

SUMMARY

The SAKAI 4 x 4 centrifugal pump coupled to YANMAR TF70L diesel engine was tested for performance and suction condition. It was tested at the recommended full throttle speed setting of the engine. The pump and engine used double groove B-section pulleys with actual diameter of 101.6 mm (4.0 inches) and 76 mm (3.0 inches), respectively, and two pieces of V-belts. The pump and engine were mounted on a steel base.

The Maximum Total Head (H) was 16.7 m at zero discharge condition. The shaft speed of the pump and engine at this point were 1954 and 2526 rpm, respectively. The Engine Fuel Consumption Rate at this point was 0.97 L/h. The Maximum Discharge Capacity of 33.5 L/s occurred at 10.1 m Total Head with Fuel Consumption Rate of 1.96 L/h. The shaft speed of the pump and engine at this point were 1646 and 2186 rpm, respectively. The Maximum System Efficiency of 18.9% occurred at 20.1 L/s Discharge Capacity and 16.0 m Total Head.

The suction condition of the pump was also tested. With zero discharge, the Maximum Total Suction Lift was 8.06 m. The Net Positive Suction Head Available at this point was 1.81 m.

PURPOSE AND SCOPE OF TEST

The purpose of the test was to establish the pump performance which included the Fuel Consumption Rate and the Total Head at varying discharge capacities and suction conditions. The pump was mounted on the laboratory test set-up shown in **Figure 1**.

The test was conducted on June 5, 2013 at the AMTEC Test Laboratory, UPLB, College, Laguna.

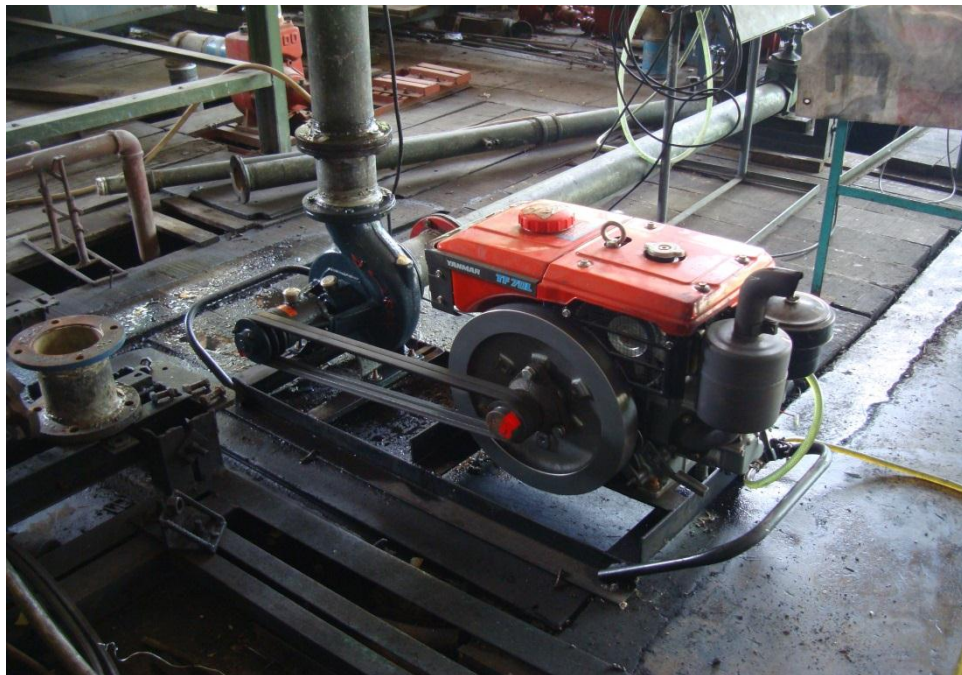


Figure 1. SAKAI 4 x 4 Centrifugal pump coupled to YANMAR TF70L Diesel engine on the test set-up

METHOD OF TEST

The pumpset was tested following the “Philippine Agricultural Engineering Standards, PAES 115:2000, Agricultural Machinery – Centrifugal, Mixed-Flow, Axial-Flow Water Pumps – Methods of Test”.

DESCRIPTION OF THE PUMP

The SAKAI 4 x 4 Centrifugal pump shown in **Figure 2**, was a single-end suction, 100 mm x 100 mm, centrifugal pump coupled to a 5.22 kW (7.0 Hp) YANMAR TF70L diesel engine. The pump and engine were mounted on a steel base. The power from the engine was transmitted to the pump shaft through a double groove B-section pulley and V-belts. The actual diameter of the pump and engine pulleys were 101.6 mm (4.0 inches) and 76 mm (3.0 inches), respectively. The suction inlet and discharge outlet were threaded and designed for a 100 mm diameter pipe. The impeller was a closed type made of brass material connected to the pump shaft through a key. It was housed in a cast iron volute casing. The priming inlet was located on top of the pump beside the discharge outlet. The discharge outlet was located on top of the pump facing upward. The stuffing box used a packing seal. The pump shaft rotated on ball bearing kept in place by a bearing housing bolted to the casing.

The primemover, a YANMAR TF70L Diesel engine, was a 5.22 kW (7.0 hp), four-stroke, water-cooled, single horizontal cylinder diesel engine. Fuel feed system was of gravity type with the fuel tank located on top of the engine. It utilized oil bath type air cleaner. Lubrication system was force-feed type. Cooling was done through the circulation of water from the cylinder jacket to the radiator. It was started by a hand crank system. A muffler was used in the exhaust system.

Table 1 shows the detailed specifications of the pump.



Figure 2. SAKAI 4 x 4 Centrifugal Pump coupled to a YANMAR TF70L Diesel Engine

PUMPSET SPECIFICATIONS

Table 1. Detailed specifications of the SAKAI 4 x 4 Centrifugal Pump coupled to YANMAR TF70L Diesel Engine

1.	Brand	:	SAKAI
2.	Model	:	4 x 4
3.	Serial No.	:	No data
4.	Make	:	Made in China
5.	Type	:	Single-end suction, self priming Centrifugal pump
6.	Size (mm)	:	100 x 100
7.	Overall Dimension and Weight (Including mounting frame and engine)		
7.1	Length (mm)	:	1320
7.2	Width (mm)	:	460
7.3	Height (mm)	:	550
7.4	Weight (kg)	:	128.0
8.	Impeller		
8.1	Type	:	Closed
8.2	Diameter (mm)	:	176
8.3	Width (mm)	:	29
8.4	No. of vanes	:	8
8.5	Material	:	Brass
9.	Suction Inlet		
9.1	Type	:	Threaded flange
9.2	Diameter (mm)	:	100
9.3	Material	:	Cast Iron
10.	Discharge Outlet		
10.1	Type	:	Threaded flange
10.2	Diameter (mm)	:	100
10.3	Material	:	Cast Iron

Table 1. (Cont'n)

11.	Casing		
11.1	Type	:	Volute
11.2	Material	:	Cast Iron
12	Stuffing box	:	Packing Seal
13.	Rotation	:	Clockwise when facing the suction inlet
14.	Rated speed (rpm)	:	No data
15.	Rated Maximum Discharge Capacity (L/s)	:	No data
16.	Rated Maximum Total Head (m)	:	No data
17.	Primemover (based on Plate Rating)		
17.1	Brand	:	YANMAR
17.2	Model	:	TF70L
17.3	Serial No.	:	DC7693
17.4	Make	:	Indonesia
17.5	Type	:	Single cylinder, 4-stroke, Water-cooled, single horizontal diesel engine
17.6	Rated Output Power (kW)		
	Maximum	:	5.22 @ 2400 rpm
	Continuous	:	4.48 @ 2200 rpm
17.8	Bore x Stroke (mm)	:	78 x 80
17.9	Displacement Volume (cc)	:	382
17.10	Cooling System	:	Water-cooled
17.11	Lubricating System	:	Force-feed
17.12	Starting System	:	Rope-recoil
17.13	Air Cleaner	:	Oil bath type
17.14	Exhaust System	:	Muffler

PERFORMANCE TEST

Results of performance test are shown in **Table 2** and plotted in **Figure 3**.

Pumpset Tested : SAKAI 4 x 4 Centrifugal pump
Coupled to YANMAR TF70L
diesel engine

Date of Test : June 5, 2013

Test Condition

Ambient Temperature (°C) : 31.8
Atmospheric Pressure (mb) : 1013
Pumping Water Temperature (°C) : 33.0

Table 2. Results of Performance Test

ENGINE SHAFT SPEED (rpm)	PUMP SHAFT SPEED (rpm)	FUEL CONSUMP- TION (L/h)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)	WATER POWER (kW)	INPUT POWER (kW)	SYSTEM EFFICIENCY (%)
2526	1954	0.97	16.7	0	0	10.0	0
2497	1910	1.42	16.9	13.2	2.18	14.8	14.8
2492	1894	1.51	16.6	17.8	2.88	15.7	18.4
2463	1869	1.61	16.0	20.1	3.16	16.7	18.9
2427	1839	1.71	15.3	21.7	3.26	17.8	18.3
2365	1785	1.83	14.3	24.6	3.44	19.0	18.1
2307	1740	1.89	13.2	27.2	3.52	19.7	17.9
2256	1699	1.93	12.1	29.4	3.49	20.1	17.4
2211	1665	1.95	10.9	31.9	3.40	20.3	16.7
2186	1646	1.96	10.1	33.5	3.32	20.4	16.2

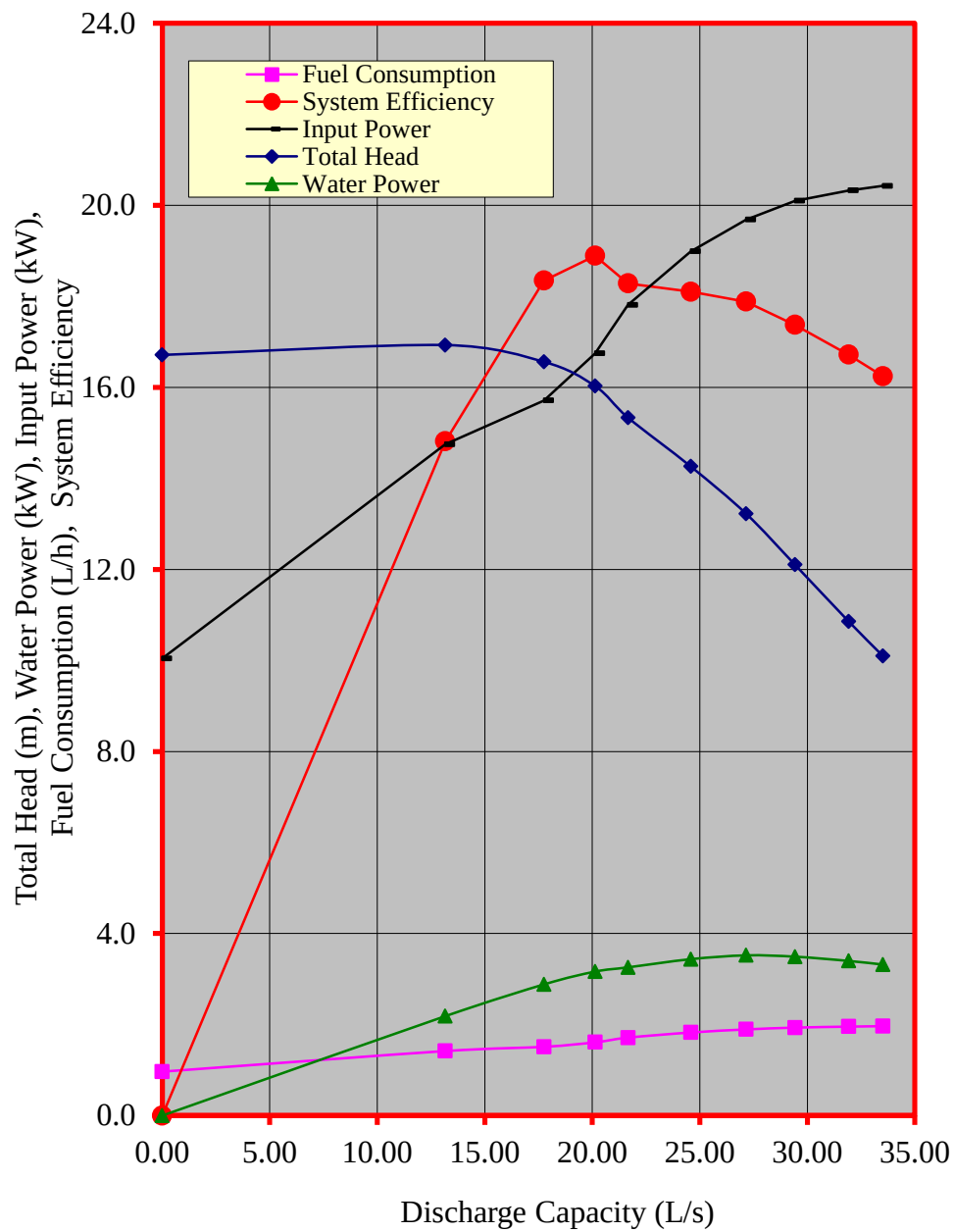


Figure 2. Performance characteristic curves of the SAKAI 4 x 4 centrifugal pump coupled to YANMAR TF70L diesel engine

CAVITATION TEST

Results of Cavitation Test are shown in **Table 3**.

Pumpset Tested : SAKAI 4 x 4 Centrifugal pump coupled to YANMAR TF70L diesel engine

Date of Test : June 5, 2013

Test Condition

Ambient Temperature (°C) : 32.6

Atmospheric Pressure (mb) : 1013

Pumping Water Temperature (°C) : 34.5

Table 3. Results of Cavitation Test

ENGINE SHAFT SPEED (rpm)	PUMP SHAFT SPEED (rpm)	TOTAL SUCTION LIFT (m)	NET POSITIVE SUCTION HEAD AVAILABLE (m)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)
2205	1658	1.99	7.88	12.3	34.2
2208	1663	2.46	7.41	12.7	33.0
2243	1686	3.93	5.94	13.5	30.8
2283	1725	5.40	4.47	14.3	28.4
2351	1779	6.17	3.70	14.6	26.2
2498	1916	7.29	2.58	13.1	10.6
2516	945	8.06	1.81	13.0	0

DATA COMPUTATION

ANNEX 1. Formula used during calculation and testing

1. Total Head (m), H

$$TDH = H_d - H_s$$

Where: $H_d = \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + Z_2$

$$H_s = \frac{P_1}{\gamma} + \frac{V_1^2}{2g} + Z_1$$

H_d = Total Discharge Head (m)

H_s = Total Suction Lift/Head (m)

$\frac{P_2}{\gamma}$ = Pressure gage reading on the discharge pipe at
gage connection converted to meter

$\frac{P_1}{\gamma}$ = Vacuum gage reading on the suction pipe at the
point of gage connection converted to meter

$\frac{V_2^2}{2g}$ = Velocity head on the discharge pipe converted
to meter. Computed by determining the velocity
of liquid on the pipe by formula, $V = Q/A$ where
 V , velocity of liquid inside the pipe in meter per
per second, Q , Discharge rate of the pump in
liters per second, and A , Cross-sectional area
of the pipe in square meter.

$\frac{V_1^2}{2g}$ = Velocity head on the suction pipe converted to
meter

Annex (Cont'n)

Z_2 = The distance from the center of the pressure gage face to the centerline of the pump in meters

Z_1 = The distance from the center of the vacuum gage face to the centerline of the pump in meters

2. Discharge Capacity, Q, in liters per second (L/s) is determined through gravimetric method using a platform scale and a stopwatch.
3. Water Power, WP, is the output power of the pump in kilowatt (kW) and computed using:

$$WP = \frac{H \times Q}{102}$$

4. Input Power , IP, is the power on the input shaft of the pump. This is the power in kilowatt required to drive the pump and is determine by the formula:

$$IP = \frac{T \times N}{974}$$

Where:

T = Torque in kilogram-meter, on the input shaft of the pump and determined by a dynamometer

N = Input shaft angular speed in RPM

5. Pump Efficiency, η ,

$$\eta = \frac{WP}{IP}$$

6. Net Positive Suction Head Available, NPSHA. This is the net force (converted to meter) needed to push the water inside the pipe and computed by the formula:

$$NPSHA = P_{atm} - P_{vp} - H_s$$

Where:

P_{atm} = Atmospheric pressure during the test converted to meters

P_{vp} = Vapor pressure of the test liquid at certain temperature converted to meters

Test Engineer:

ROMULO E. EUSEBIO

Checked by:

DARWIN C. ARANGUREN

Approved for release:

DELFIN C. SUMINISTRADO

Director

Date

N.B.

This test report should not be used for advertisement or for promotion of the product but only for evaluation purposes and should not be reproduced in whole or in part without the permission of AMTEC.